

REVIEW

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Entrepreneurial strategies for sustainable growth: a deep dive into cloud-native technology and its applications

T. A. Alka¹ , Aswathy Sreenivasan² and M. Suresh^{1*}

Abstract

The study highlights the impact of cloud-native technology in entrepreneurship for innovative business models, scalability, and cost efficiency by combining the trends including microservices, serverless computing, and technology disruptions. The study attempts to examine the role of cloud-native technology affects the entrepreneurial landscape in creating and advancing innovative business models and approaches, and the difficulties while applying cloud-native technology. The systematic literature review based on PRISMA is used to explore cloud-native technology and its applications by analyzing selected 111 studies from Scopus, Web of Science, Dimensions, and ProQuest. The study identified that one of the main forces behind disruption is the quick development of new technologies including blockchain, 5G, IoT, quantum computing, and AI. Cloud-native technology supports innovative business models in entrepreneurship such as SaaS, serverless computing, managed Kubernetes, data analytics platforms, BaaS, and Edge Computing. The urgency to incorporate and develop new security features and tools for measurement is necessary. The study shows that startups can use cloud-native technologies for innovative business models that rely on serverless computing, subscription services, and on-demand cloud services which are providing opportunities. The findings of the study highlight the need for entrepreneurs to utilize emerging technologies to meet competition, and there is a requirement for continuous investment in data analytics, automation, cloud solutions, and emerging technologies. Collaboration is essential for the application of the technologies for early stage startups and enterprises. Incubators, accelerators, and venture capital firms support essential to the adoption of disruptive technologies. The policymaker's and enterprises' focus and actionable intervention are necessary for developing mechanisms to solve the barriers of regulatory compliance, lack of skilled manpower, data security, privacy issues, and upfront investment for entrepreneurial success.

Keywords Entrepreneurship, Technology disruption, Cloud-native technology, Business models, Microservices, Serverless computing

Introduction

Technology disruption creates opportunities as well as threats to the global entrepreneurial environment. The major component of entrepreneurship is the creation of innovative ideas to survive and grow in the international competitive environment. The technological revolution in the 21st century, such as “big data analysis,” “artificial intelligence (AI),” “fintech,” and “cloud computing,” creates the necessity of technology in life. Likewise, the role of innovative technology in entrepreneurship is also

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inevitable. The conventional form of selling and buying products heavily depends on the classic point of sale idea changing [1]. It was changed to online buying and selling. Recently, different online shopping websites and applications, etc., have evolved for conducting online selling. Technical aspects of cloud-native technologies, such as serverless computing, Kubernetes-like orchestration tools, containerization, and microservices architecture, play important roles in entrepreneurship. The development of technology over the recent decade has rapid growth. Cloud-native technology is creating disruptions and fostering innovation in different sectors. The importance of cloud-native technology in the 5th Generation networks is crucial which aims for agility and innovations [2]. The technologies of containerization including Docker and orchestration systems such as Kubernetes are playing a role in the infrastructure of cloud-native having impacts on the high storage capacity challengers as well as enhancing flexibility [3]. For example, the Dao-Cloud development of the microservice governance management of containers, and also the more than one cud orchestration shows the digital transformation impacts in the enterprises [4]. The Kubernetes-supported storage facilities accelerate the scalability as well as the solutions to the challenges of conventional storage. Kubernetes has an automated toolchain and sophisticated features that promote storage [3]. Event-driven microservices help satisfy the satisfying of service level objectives (SLOs) [5]. The cloud-native services enabled by the IoT enhance the high throughput connectivity and the less latency when multiple clouds in different areas [6]. Likewise, edge computing also ensures the fast delivery of services and the boosted performance of the system. For example, the observability in the cloud native and the Log management platforms are developed for the effective management of the difficulties, and log information [7, 8]. The Cloud-Native Technology must be “IDEAL,” in which “I” stands for Isolation, “D” stands for Distribution, “E” denotes Elastic, “A” represents Automated Management, and finally, “L” denotes Loosely Coupled. The factors influencing the adoption or use of cloud-native technology are speed, safety, diversity, etc. [9]. “The emergence of Modern Big Data technology leads to increased data size in daily life, and it results in the need for efficient large data analysis” [10]. Cloud-native technology is not a new topic. According to Kovács [11], “Cloud-Native Applications (CNAs) denote a graph of containers for communicating microservices.” Kubernetes and Apache Mesos are the most popular open-source container orchestration tools. There are different paper publications, in which authors from academic and industry fields define cloud-native applications [9, 12]. The proper prediction leads to improving resource utilization and service levels with no

objection [13]. However, containers that assist different sorts of businesses have different perceptions and needs on resource requirements; thus, the prediction methods via one general model cannot work for all types of containers. Cloud technology is also used in public health services as a machinery learning platform that merges machine-learning frameworks and cloud-native technology [14]. Difficulties experienced in traditional command and control systems (closed systems), the inefficiency of man-machine collaboration, and poor scalability are the reasons for cloud-native technology’s application—method to create an open command and control system. Cloud technology-based developed systems consist of two main parts, the diagnostic knowledge base and the inference method. Cloud-native technology is related to technology disruption.

Aim

While reviewing existing studies, they do not focus on all the business models together related to cloud-native technology, and also they are looking at the side of the advantages and disadvantages. The ethical considerations associated with cloud-native technology are also an underexplored area. A thorough investigation and analysis of the entrepreneurial strategies companies use to achieve sustainable growth, focusing on the effects of technological disruption and the adoption of cloud-native technology in the modern business environment. This analysis intends to offer insightful perspectives into the changing entrepreneurial landscape by highlighting best practices and new trends to help entrepreneurs and organizations pursue long-term sustainability and competitiveness. This study aims to understand how entrepreneurs may use these technologies to design, launch, and scale their applications more quickly and affordably. In today’s changing economic environment, studying cloud-native technology’s role and effects on entrepreneurship and entrepreneurial activities has potential and significance. A paradigm shift in corporate operations has been brought about by cloud-native technology, enabling startups and entrepreneurs to use scalable and affordable computing resources without making sound and significant infrastructure investments.

Research questions

This systematic literature review examines the existing published works and synthesizes the literature on entrepreneurial strategies for growth with the help of cloud-native technology and technology disruption. The study deeply dives into the characteristics, impact, and challenges of technology disruption and cloud-native technology in entrepreneurship. The major four research questions addressed in this study are;

How do disruptive technologies like cloud-native technology affect the entrepreneurial landscape in creating and advancing innovative business models?
 What are the cloud-native technology business models and approaches applicable in entrepreneurship?
 What are the difficulties experienced while applying disruptive technologies and cloud-native technology?
 How the ethical and regulatory issues be solved?

The first research question mainly covers the impact of technology disruption including cloud-native technology, especially looking from the perspective of low-end disruption, opportunities, and how they are contributing to novel innovation. This will help to understand the favorable or negative impact on the entrepreneurs and it has a wider scope for developing entrepreneurial strategies. It also deals with understanding forces that lead to technology disruption. This also examines the role of the development of technology including the development and growth of AI and blockchain. The exploration of the forces helps entrepreneurs to know the influence and the ability to meet the customer requirements. The cloud-native technology-enabled business models are assessed in the second research question. The model's SaaS, BaaS, etc., are examined here. This helps in the identification of various models that can accelerate innovation. The challenges of cloud-native technology and technology disruption integration into enterprises from the finance, regulatory, and human resource perspective are evaluated in the last research question. The identification of the challenges will inform the entrepreneurs, policymakers, and supporting mechanisms to make actionable interventions. The study also examines recommendations for how ethical and regulatory issues such as data privacy, security, and compliance issues can be effectively overcome. All the research questions addressed the impact of technology trends on entrepreneurial operations.

Scope and significance of the study

Cloud-native technology is crucial for digital transformation in firms which plays a role in innovation, and adaptation, competitive strength to the transitions happening in the digital world to be resilient [15, 16]. They offer unique opportunities to the enterprise. The Startups always focusing on coming up with new technology get benefits due to the "pay as you go" model, enhance cost efficiency and elimination of upfront investment, and also foster resource efficiency. Startups, especially small-scale firms, have opportunities for survival with cloud-native technology. The enterprises can scale up their forms move toward the demand in the market and improve operations. The synergy between the cloud-native and the technology trends such as IoT and big data

will improve the competitive advantage in the operation with real-time and stimulate innovation [17]. Similar to startups, large business endeavors are adopting cloud-native for innovations, cost efficiency, and marketing opportunities resulting in scalability and growth, and also facing competition. Due to the significance of the cloud native which offers large opportunities for entrepreneurship including startups. Hence, the research on examination of the impact and role is essential. The importance of cloud-native technology in entrepreneurship is relevant and most significant due to the transformation in the business environment, especially the technology disruption. The cloud-native platforms promote agility and entrepreneurial success even though face stringent competition, drastic innovations, and dynamic digital trends. Nowadays, the recognition of the advantages of utilizing digital trends has increased. The cloud-native is also a prominent opportunity recognized by entrepreneurs. The exploration of the role of cloud-native technology in entrepreneurship endeavors will offer insights, expand the existing knowledge, and provide future research avenues. The study shows the current state of the art in terms of the emerging trends, patterns, evolution, and challenges, that lead to the call for the actionable intervention of the practitioners, and policyholders to frame suitable policies, programs, practices which ensures the adoption of cloud-native and overcoming problems of the entrepreneurs. The systematic literature review offers a comprehensive and deeper analysis with adequate supporting evidence to present the current trends and reveal the gaps, building for further research systematically. The cloud-native across different industries, especially cloud-native systems, such as microservices, containerization, and serverless computing is inevitable in the business strategy and identification of best practices. The exploration will fill the research gap through a comprehensive examination of the what is the role of these systems in entrepreneurship. Similarly, the systematic review is essential for a holistic understanding of the cloud-native phenomenon reshaping the entrepreneurial landscape. This study explores how cloud-native approaches can allow entrepreneurs to innovate more quickly, streamline development processes, and improve the agility of their businesses. Cloud-native technology also creates new opportunities for start-up businesses to grow and compete in various sectors by easing operational complexity and promoting quick application deployment. Aspiring and experienced entrepreneurs can benefit greatly from understanding the implications and best practices of cloud-native adoption, making it easier to exploit their respective industries' unique strengths. In today's quickly changing business environment, the study on the impact of technology disruption on entrepreneurship has scope

and impact. Technology is developing at an unpredictable rate, changing how entrepreneurs establish and scale their businesses. This study can provide insight into how cutting-edge technology transforms market dynamics, consumer behavior, and company structures. It can reveal tactics for utilizing disruptive technology to build competitive, inventive, and sustainable firms. This study examines how entrepreneurship is changing in the modern day concerning digital transformation. Its main goal is to highlight how entrepreneurs may use disruptive and cloud-native technology to achieve sustainable growth in a dynamic industry environment.

The study attempts to identify the role of cloud-native technologies and the technology disruptions in entrepreneurship. The study offers the idea that startups can use cloud-native technologies for innovative business models that rely on serverless computing, subscription services, and on-demand cloud services which are providing opportunities. The findings of the study highlight the need for entrepreneurs to utilize emerging technologies to meet competition, and there is a requirement for continuous investment in data analytics, automation, cloud solutions, and emerging technologies such as biotechnology, 5G, IoT, quantum computing, and AI. Collaboration is essential for the application of the technologies for early stage startups and enterprises. Incubators, accelerators, and venture capital firms support essential to the adoption of disruptive technologies. The mechanisms to solve the barriers of regulatory compliance, lack of skilled manpower, data security, privacy issues, and upfront investment are essential for entrepreneurial success. The policymakers and enterprises focus and actionable intervention are essential. The adoption of containerization and Kubernetes helps in the scalability and adaptability.

The study is organized as follows; the first section "Introduction" explains the purpose, relevance, major research questions, scope and importance, and also how the study is contributing. The second part "Literature review" shows the background and key concepts related to cloud-native technology and technology disruptions. The "Methodology" explains the procedure of PRISMA, the exclusion and inclusion criteria followed in this research, and also the possibility for the replicability and the reproducibility of this research. The results section shows the systematic review of existing studies to answer research questions. Discussion summarizes the key findings and implications followed by Recommendations and conclusion to the study.

Literature review

Understanding cloud-native technologies

The National Institute of Standards and Technology (NIST) definition of cloud computing defined

three fundamental and widely used service categories: "Infrastructure as a Service" (IaaS), "Platform as a Service" (PaaS), and "Software as a Service" (SaaS). "IaaS offers the greatest flexibility for any consumer product, but it almost completely masks the infrastructure's operational complexity." "SaaS, conversely, virtually completely conceals operational complexity but is inadequate for many cases involving software developed by end users." "PaaS is a compromise that allows user-created software to operate conveniently while maintaining operational complexity but at the cost of accepting platform-related lock-in to some extent" [18]. Cloud-native technology helps in efficiency as well and performance technology is the application that mainly deals with the operation in the cloud environment consisting of diverse characteristics [19]. According to Kosińska et al. [20], "a microservice can be deployed, scaled, and restarted without disturbing the work of other microservices. They communicate with the environment through the HTTP API. Microservices decompose monolithic applications into small, independent components." The next basic component of cloud-native technology is a container that can pack the microservices according to specific requirements [21]. Digital transformation is the transformation that happened recently, which focuses on the agile innovations able to satisfy the market requirements through new novel innovative technology trends and they have transformed the operations [22]. The cloud-native technology is mainly driven by the growing importance of DevOps which accelerated the development of cloud-native technology [23]. DevOps integrates the development of software and information technology operations which enhances continuous and prompt service delivery with less cycle time [21]. The edge computing continuum is incorporating cloud-native technologies with edge computing enriching the processing with low latency and increasing the efficiency of the IoT [24]. The technology disruption contributed by the cloud-native technology creates agile and adaptive systems that are capable of differing the influence of the emergence and the growth of the internet during the 1990s. According to [25], "The growth of cloud-native computing is motivated by the development of cloud computing delivery mechanisms." The groundbreaking move toward cloud computing is due to its agility, scalability, and resilience. Microservices have rapidly acquired popularity in the Information Technology (IT) industry. According to Weerasinghe and Perera [26], "cloud-native" is derived from cloud computing technologies and refers to the development and deployment of applications in a cloud environment.

Microservices are widely used to deploy applications to improve management systems' flexibility and agility. However, when implementing cloud-native applications, serverless computing has grown in popularity. In contrast to microservices, serverless architecture offloads operational maintenance and management of the underlying infrastructure to the cloud provider. Users limit their attention to developing business logic. With both technologies, applications can benefit from quicker delivery, lightweight scalability, and lower development and maintenance costs. Arguments exist over which application deployment strategy should be used. Network Function Virtualization (NFV) plays a role in network slicing in business. Microservices are recently launched and de-registered in cloud-native systems, and standard security assessment tools have difficulty finding them. As a result, security assessments in these settings demand novel strategies specifically tailored to and integrated with Cloud-Native Technology [27]. Using varied robot software packages that already exist and put them on the cloud without making any further adjustments thanks to the Robot Operating System (ROS). Robotic applications can be detached from the cloud using microservices design and container technologies [28]. Two service modeling tools are available for building cloud applications and services. It offers collections of preconfigured software assets and the connections among them that provide information on how to install and configure particular services in the cloud properly. The other tool is Fabric8, a Kubernetes and Docker platform for orchestration and virtualization, respectively. Microservice development, deployment, and continuous integration are supported. However, these two service modeling tools do not enable infrastructure planning or provisioning or particular provisioning for time-critical applications. Technology disruption has been driven by cloud-native technology, which allows for the quick development and deployment of novel solutions and is defined by cloud-based infrastructure, microservices architecture, and DevOps practices. Offering subscription-based models, in which clients pay a monthly or yearly price to use your product or service, is simple with cloud-native technology. This can be accomplished by building a devoted customer base and generating recurring revenue. Freemium models are based on cloud-native technology, in which customers can avail of the basic version of products or services free of cost. However, there is a subscription for additional features or capabilities. This is applied effectively in the case of new clients and their retention in the coming years. The clients pay for products or services, depending on how much

they use them, which are simple to offer thanks to cloud-native technologies. Table 1 and Fig 1 show the concepts related to cloud-native technology from basic to advanced or complex elements.

Understanding technology disruption in entrepreneurship

According to Merriam-Webster Dictionary, 2020, "disruptive is derived from "disruption," which is defined as "a break or interruption in the normal course or continuation of some activity, process, etc." [36]. Particularly in the last two decades, technology has advanced quickly thanks to ground-breaking and disruptive technologies that continue to question and modify "traditional" methods of doing things [37]. For entrepreneurs, the speed of technological disruption and innovation is profoundly impacting almost all value chains and producing long-term improvements in how organizations are run and sustained on entrepreneurship innovation. However, low-end disruption has a favorable but minor impact [38]. For entrepreneurs, the speed of technological disruption and innovation is profoundly impacting almost all value chains and producing long-term improvements in how organizations are run and sustained. According to various researchers, "Disruptive innovation is not merely an outcome, but a complete and progressive process" [39–41]. Disruptions in technology and the economy, along with new opportunities and dangers to the models of global entrepreneurship, have changed cultures all over the world. "Analysis of entrepreneurship as a dynamic phenomenon focuses on how it can be used to manage risk associated with market uncertainty, innovation, competitiveness, and restructuring, as well as to produce new knowledge" [42]. "Three areas of opportunity for digital entrepreneurs have been created due to the traditional components of the industry's value delivery being phased out by technological shocks: Manufacturing, distribution, and transaction" [43]. "By releasing new items, enhancing the usability of existing products, and creating a digital connection between them, digital disruptors can use digital platforms" [44]. By introducing goods and services that are radically cheaper, better, and more convenient, disruptive technologies either increase the industries they enter or start wholly new ones. Due to their ability to let people with limited technological sophistication join the workforce and compete, these disruptive technologies frequently hurt how people participate in the labor force.

Methodology

The meta-analysis extracts the quantitative information required to combine many studies statistically [45]. Likewise, the systematic literature review synthesizes existing literature. The Preferred Reporting Items for Systematic

Table 1 Overview of the concepts in cloud-native technology

Concepts	Meaning	References
Cloud-based applications	This application helps in horizontal and vertical scaling with automation integrating with developed tools capable of the transition from the very natural to the innovative recent applications. E.g., Dropbox	Nascimento et al. [29]
Cloud-Native Application Bundles (CVAB)	The open-source packaging solutions CNAB provide the proper management of the cloud-native technology life cycle from the installation to the exclusion or removal and also they help in the expansion of the multi-cloud services and operations expansions. E.g., Duffle	Nascimento et al. [29]
Containers	They are packaging the software into elements and it is the basic component of the cloud-native technology. E.g., Docker	Fu et al. [30], Kosińska et al. [20]
Cluster, Master, Worker	The containers and orchestrator perform in the cluster having more than two nodes showing the individual computing source with the ability to integrate all the tasks. Among these nodes, the master is the node that handles the workload managing nodes, especially considering the nodes that ferry workloads as workers.	Kosińska et al. [20]
Orchestration	The tools that are capable of effective handling of containerized applications, and the deployment, contain lifecycle management. E.g., Kubernetes	Paul [31]
Edge native	The principles of cloud-native technology application are mainly design-related which is adopted in edge computing devices.	Vaño et al. [32]
Service-Oriented Architecture (SOA)	SOA can satisfy the enterprise requirements in real-time, flexibility, and developing competitive advantage by the functioning of the resources across the network and has the features of modularization, encapsulation, and loose coupling of early distributed computing methods.	Su et al. [33]
Service mesh	Service mesh is capable of managing the communication and networking among the microservices which enhances the fast of microservices with the security, handling of traffic, and the opportunity to watch the microservices. E.g., Istio	Pallewatta and Babar [34]
Microservices	The microservices are the foundational element in the cloud-native technology which is an emphasis division into the particular segment able to offer independent services and its scalability.	Kim et al. [35], Bharadwaj et al. [21]
Serverless computing	Serverless computing mainly comes into the picture in the role of managing the infrastructure and there is only a requirement for the developer to develop codes. Hence, this promotes the scaling and cost-less facility. E.g., AWS Lambda, Google Cloud Functions, and Azure Functions	Nascimento et al. [29]

Reviews and Meta-Analyses (PRISMA) flow is suitable for systematic literature review. The screening procedure is visually summarized in the PRISMA flow diagram [46]. The number of papers found is first noted, and then, the selection process is made explicit by reporting on choices made throughout the systematic review. At various phases, the number of articles is kept track of [47]. Scopus, Dimensions, Web of Science, and ProQuest databases were taken to search for documents related to this theme without any time restriction. A total of 189 papers were found by comprehensive search on this theme. Out of the initially identified papers, 111 fulfilled the specific eligibility criteria.

Raman et al. [48] highlighted that the PRISMA protocol helps in the acquisition of a less biased set of data that

contains inputs for a systematic literature review. By following PRISMA, comprehensive data analysis is possible through the relevant dataset which is undergoing through major 4 stages of purification systematically. It also promotes the reproducibility of the data [49, 50]. According to Nedungadi et al. [51], “the PRISMA framework is a recognized standard for conducting evidence-based systematic reviews and meta-analyses. The PRISMA statement consists of four essential phases: identification, screening, eligibility, and inclusion.”

The four steps in the PRISMA are;

- (1) Introduction
- (2) Screening
- (3) Eligibility

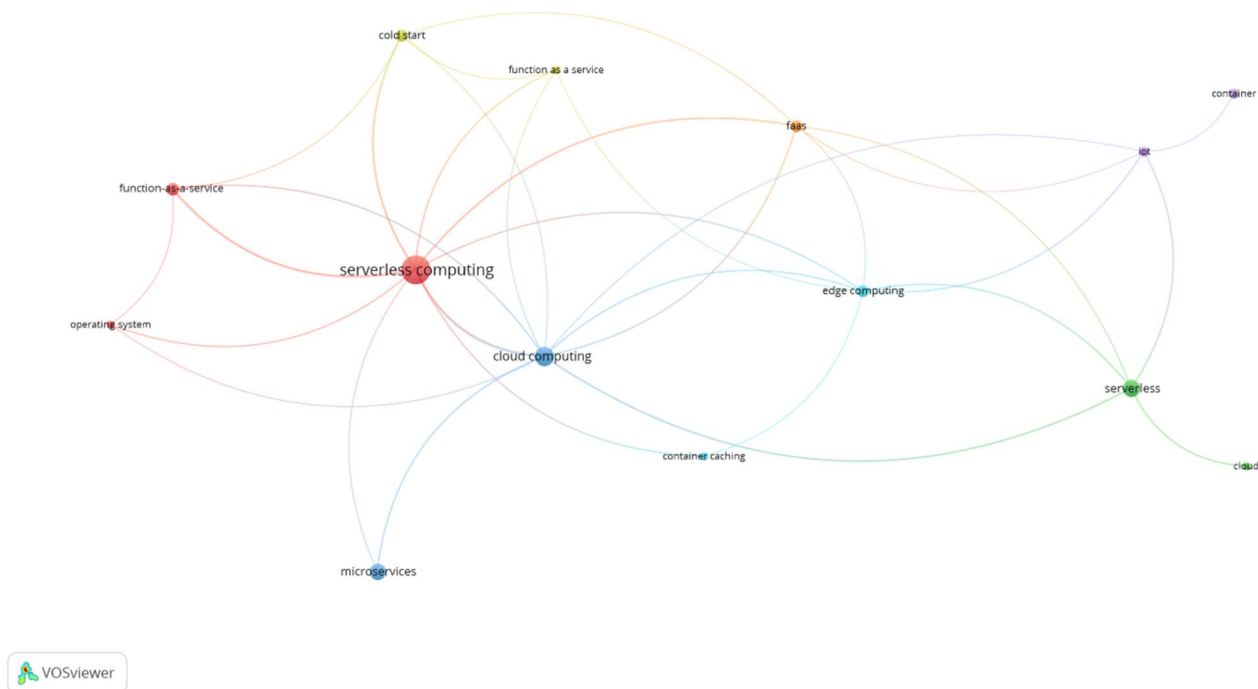


Fig. 1 Concepts in cloud-native technology

(4) Included

The detailed inclusion and exclusion criteria are the following in the four stages followed in the PRISMA framework;

(1) Introduction

Search database: A basic search was carried out during this initial stage. The search is conducted in the most prominent databases such as Scopus, Web of Science, Dimensions, and ProQuest. The first document was published in Dimensions in 2013. The choice of database is most crucial while doing a systematic literature review. Here, the most suitable coverage databases are selected. Scopus is selected due to the highly indexed database, large coverage in different disciplines, availability of Boolean operators, and more peer-reviewed and quality papers are available. Advanced document search and filtration are possible during the search [48–50, 52]. The next database is the Web of Science (WoS). According to Alka et al. [53], “Web of Science (WoS) is a multidisciplinary research database covering the “Science Citation Index Expanded” (SCIE), “Social Sciences Citation Index” (SSCI), “Arts & Humanities Citation Index” (AHCI), and “Emerging Sources Citation Index” (ESCI).” The feature of high-quality publications and coverage of interdisciplinary studies which is most beneficial for

conducting a study that deals with technology trends are the driving factors for choosing WoS. Dimensions have a large number of papers by combining papers from different sources. The broader coverage and a open access papers availability are the advantages of choosing dimensions. The publications of allied research and industry-related practical application publications are essential for exploring cloud-native technologies. The documents found from the initial search are 164 (Scopus: 92; WoS:1; Dimensions: 60; ProQuest:11) without any exclusion criteria.

Search keywords: The major keywords used are: ("cloud native technology" AND "entrepreneurship") OR ("serverless computing" AND "entrepreneurship") OR ("technology disruptions" AND "entrepreneurship") OR ("cloud native technology" AND "startups") OR ("microservices " AND "startups") OR ("serverless computing" AND "startups") OR ("technology disruptions" AND "startups"). This keyword comprehensively covers the literature on cloud native technology and entrepreneurship and also startups.

Search period: The study aims to map the recent literature, especially during the current decade, from 2015 to 2024. The reasons behind search limiting are: the goal to discuss recent and up-to-date, influential research to reveal the trends, patterns, etc. The major technological disruptions and trends have evolved and developed recently. It is better to rely on the new studies while

focusing on analyzing the digital trends and digital transformation. The studies did not adequately cover the modern technologies before 2015 and mostly while discussing cloud-native technology. After 2015 only the adoption of cloud-first or cloud-native strategies is growing. Here, one document published in Dimensions in 2013 was not selected. The selected documents are 163 (Scopus: 92; WoS:1; Dimensions: 59; ProQuest:11).

(2) Screening

Language: The documents are published mainly in one language “English.” Hence, there is no specific filtration applied here. English is highly recognized and commonly accepted, and the most used language is research and the problems in the misinterpretation of results can be reduced while depending on English. 163 documents in this phase are carried to the next filtration criteria.

Document type: The documents that are published as “monograph” and “preprint” are not taken. The study only relies on peer-reviewed papers and conference proceedings. Nine documents in the Dimensions are excluded. Among 163, the selected documents in this step are 154 (Scopus: 92; WoS:1; Dimensions: 50; ProQuest:11).

Stage of publication: The major two stages of publication are: “final” and “article in press.” In this study, only the final-stage published papers are selected for exploration. Therefore, 3 papers in Scopus published in Article in press” are removed. The number of documents eligible here is 160 (Scopus: 89; WoS:1; Dimensions: 50; ProQuest:11).

Duplication: There are no duplicated documents found.

Full-text accessibility: 122 papers are full-text accessible. Among these accessible papers, 11 documents are not suitable for the aim of this research. The limits to full text and excluding irrelevant papers are mainly for conducting rigor, comprehensive, quality, and less biased analysis. Hence, the selected number of papers is 111 during the second stage of “screening.”

(3) Eligibility

This is the third stage of PRISMA. There is no special purification involved in this process

(4) Included

The last step is the “Included.” The 111 documents are eligible and included in the analysis. The Scopus includes 87 documents, WoS comprises 1 paper, Dimensions consists of 41 publications, and the 11 from ProQuest.

The PRISMA guidelines are applied by addressing each research question. During the identification stage, we

searched the databases Scopus, Web of Science, Dimensions, and ProQuest with search terms “cloud native technology,” “entrepreneurship,” “serverless computing,” “technology disruptions,” “startups”, “microservices,” and “serverless computing.” This combination of keywords comprehensively covers the literature on cloud-native technology and entrepreneurship and also startups, which is directly linked to the studies to answer the research questions. The PRISMA standardized approach ensures the replicability of the research in the literature review. The selected databases have high coverage in various disciplines. Limiting to the search period from 2015 to 2024 is the drive for the mapping of recent trends effectively. Similarly, the inclusion and exclusion criteria in the screening phase also followed by avoiding the low-quality document types, and irrelevant themes to the research questions, and only focusing on the studies that directly related to answering the research questions are helping for a systematic review of this field. The eligibility stage is also followed with a clear intention to include the relevant studies that are helpful for each research question in the study and the accessibility of full text helps to do a comprehensive exploration. The final included 111 research addressing each of six research questions on the impact of disruptive technologies, the role of cloud-native technology in innovative business models, drivers for technological entrepreneurship, challenges in the technology disruption adoption and the effects of the technology disruption, and finally the emerging trends in the cloud-native technology. The PRISMA standardized approach ensures the replicability of the research in the literature review. The selected databases have high coverage in various disciplines. The paper is a pure systematic literature review. The main aim of this paper is to understand the existing research for identifying research gaps, synthesizing the body of knowledge, and offering future research directions by examining the holistic view of the particular theme. The PRISMA protocol is followed for the selection of documents by following inclusion and exclusion criteria. Here, no statistical models are employed. The major aim of this research is to synthesize the existing studies on this theme rather than analysis on the numerical data from different studies. The data are collected from the prominent four databases such as Scopus, Web of Science, Dimensions, and ProQuest. The data were downloaded, and the inclusion and exclusion criteria were applied to the language, document types, publication stage, relevancy, and the scope of the study. The final selection of the documents met the exclusion and inclusion criteria and also discussed the key themes and trends, and also showing the pattern of the existing body of knowledge. The VOS viewer is used to identify the occurrence of the themes in the cloud-native

technology. The narrative synthesis of the literature by understanding the concepts, themes, and findings of all selected studies individually and then arranging the study based on the research question and scope of the research. This qualitative exploration of all the selected studies is to get a current state of the art of the literature.

Results

Figure 2 shows the selection of the studies that are selected during the step-by-step process of the PRISMA protocol explained in the previous section to synthesize the results and findings of the selected studies from

four databases after meeting the inclusion and exclusion criteria.

Cloud-native technology and technology disruption in entrepreneurship

The Fourth Industrial Revolution 4.0, which is just beginning, is characterized by the merger of digital and physical technology, which presents previously unheard-of opportunities and difficulties [54]. The deregulation of technological services presents opportunities for company growth to promote “Internet-based services,” raising customer demand for cheaper and quicker Internet. Organizations have benefited from service innovation by

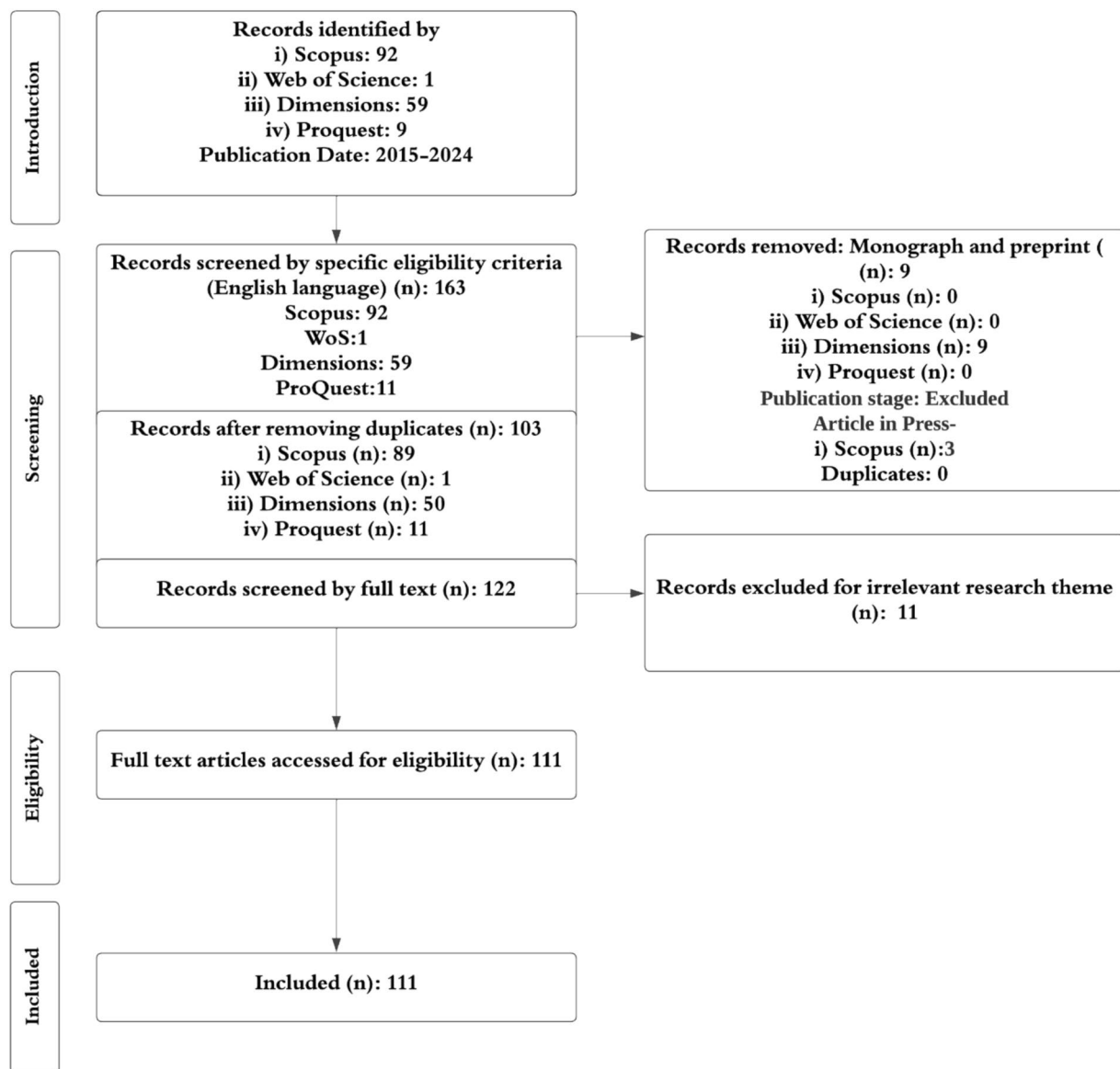


Fig. 2 Flow diagram by PRISMA model

organizing future knowledge management capabilities that are nimble and flexible in providing information and solutions and improving big data analytics. From a managerial standpoint, this research thoroughly explains how Service Innovation affects businesses, how to attain it, what factors go into it, and how to link it to performance [55]. Application development and deployment are sped up by cloud-native technology [15, 16]. Applications built for the cloud are geared toward high availability and disaster recovery. They are spread across several data centers, lowering the possibility of downtime brought on by hardware malfunctions or natural calamities. Because of their resilience, firms can carry on even in difficult situations.

Understanding technology disruption is based on the ground-breaking research of Clayton Christensen, as described in his book "The Innovator's Dilemma." According to Christensen, "Established businesses tend to concentrate on their current clients and markets, they frequently struggle to adapt to disruptive developments" [56]. Literature reviews have looked at the connection between entrepreneurship and technological disruption. According to researchers, entrepreneurs frequently bring forth disruptive breakthroughs because they are more willing to take chances and consider outlandish concepts. "The relationship between ecosystems and disruptive innovation has received some attention in several evaluations" [57]. They emphasize the value of cooperative networks, such as those between startups, financiers, and research organizations, in promoting disruptive entrepreneurship. Entrepreneurs can find gaps in existing markets and build solutions that use emerging technologies due to technology disruption. The success of businesses like Airbnb in the hospitality industry exemplifies how entrepreneurs can venture into new waters and defy industry standards [40]. Because outdated systems and procedures constrain them, established businesses frequently find it difficult to adjust to technological disruption. Entrepreneurs must prioritize sustainability by developing a culture of constant innovation and adaptability if they want to succeed in a time of technological upheaval [58]. A key component of sustainable entrepreneurship is anticipating technology changes and aggressively looking for expansion prospects. Technological disruption is key in modern entrepreneurship, altering sectors, spawning fresh business ideas, and providing chances for creative businesspeople. Entrepreneurs hoping to survive and prosper in this dynamic environment may find it crucial to embrace disruptive technologies and use cloud-native technology. Entrepreneurial disruption is when novel concepts, goods, or technologies hit the market and radically alter how companies conduct themselves in a certain sector. It frequently entails the

replacement of well-established businesses, goods, or services by newcomers who provide better, more affordable, or more creative solutions. Disruption is a key concept in entrepreneurship and corporate innovation and can have enormous economic and social ramifications. Disruption involves presenting a game-changing invention that may first target a small market but ultimately unseats seasoned companies by providing higher value to a larger audience, going beyond incremental improvements or sustaining advances. Targeting underdeveloped or ignored customer categories is usually where it begins. Disruptive innovations can completely alter whole markets, creating new market leaders while eliminating old firms. This dynamic gives newcomers the chance to question the current quo. Because of the decreased entry barriers made possible by technology, startups, and entrepreneurs can now compete with more established, larger organizations. Innovation may be encouraged by this democratization of entrepreneurship. By providing consumers with better, more reasonably priced, or more convenient goods and services, disruption helps them. It promotes competition, which results in better selections and more affordable rates. By spawning new industries and employment prospects, disruption can promote economic growth. Additionally, it promotes spending on research and development. There are no geographical bounds to disruption. Innovations have a swift global reach and can open up new business chances. Several causes and reasons that frequently modify markets, sectors, and business models are what propel technological disruption in today's entrepreneurial environment. One of the main drivers of disruption is the quick development of new technologies, including "biotechnology," "blockchain," "5G," "IoT," "quantum computing," and "AI" [59]. Using these technologies, entrepreneurs can develop novel products and business strategies to compete with market leaders. To be competitive, companies in all industries are undertaking digital transformation. New methods of conducting business and providing customer service are being made possible by transitioning from traditional to digital processes, including automation, data analytics, and cloud computing. Disruption is caused by customer choices and behavior shifts, particularly in the digital space. How entrepreneurs interact with clients and provide goods and services has changed due to the growth of e-commerce, online streaming, social media, and mobile apps. Disruption can flourish in thriving startup ecosystems with access to venture capital, accelerators, and incubators [60]. Innovative entrepreneurs can take on established players with the help of these ecosystems. Industry disruption can result from regulatory changes creating new opportunities or placing new restrictions. Entrepreneurs can benefit from these

developments if they comprehend regulatory environments and successfully manage them. The global economy's interconnection has allowed companies to develop and grow quickly across borders. Disruptive ventures have been fostered by access to international markets, talent, and resources. Due to collaborative ecosystems and open innovation approaches, entrepreneurs can access external knowledge, technologies, and resources. This can foster innovation and established business models. Sustainable and socially conscious entrepreneurship has become more popular due to growing public awareness of environmental and social challenges. Disruptive technologies substantially impact the development and adoption of novel business models [61]. It gives entrepreneurs the chance to rethink entire industries and present brand-new solutions. By utilizing cloud-native technology, entrepreneurs can capitalize on the potential of data. Advanced analytics solutions are available on cloud platforms, enabling companies to better understand consumer behavior, market trends, and operational effectiveness [62]. Entrepreneurs can make intelligent judgments

and optimize their strategy thanks to this data-driven approach. The seamless integration of cloud-native technologies with other cloud services and outside applications. This ecosystem integration streamlines company processes and enables entrepreneurs to use a variety of products and services without having to invest heavily in development. The entrepreneurial ecosystem now includes cloud-native technology and a wider range of technological disruption. While they have many benefits, they also present significant obstacles that entrepreneurs must overcome to fully realize their potential [63]. Figure 3 presents the key trends in the cloud-native technology.

Cloud-native technology approaches and business models

A growing emphasis on cybersecurity within cloud-native architectures, increased adoption of serverless computing and Kubernetes, and the development of environmentally friendly cloud-native solutions are the most recent trends in entrepreneurs' use of cloud-native technology [64]. Serverless technologies, such

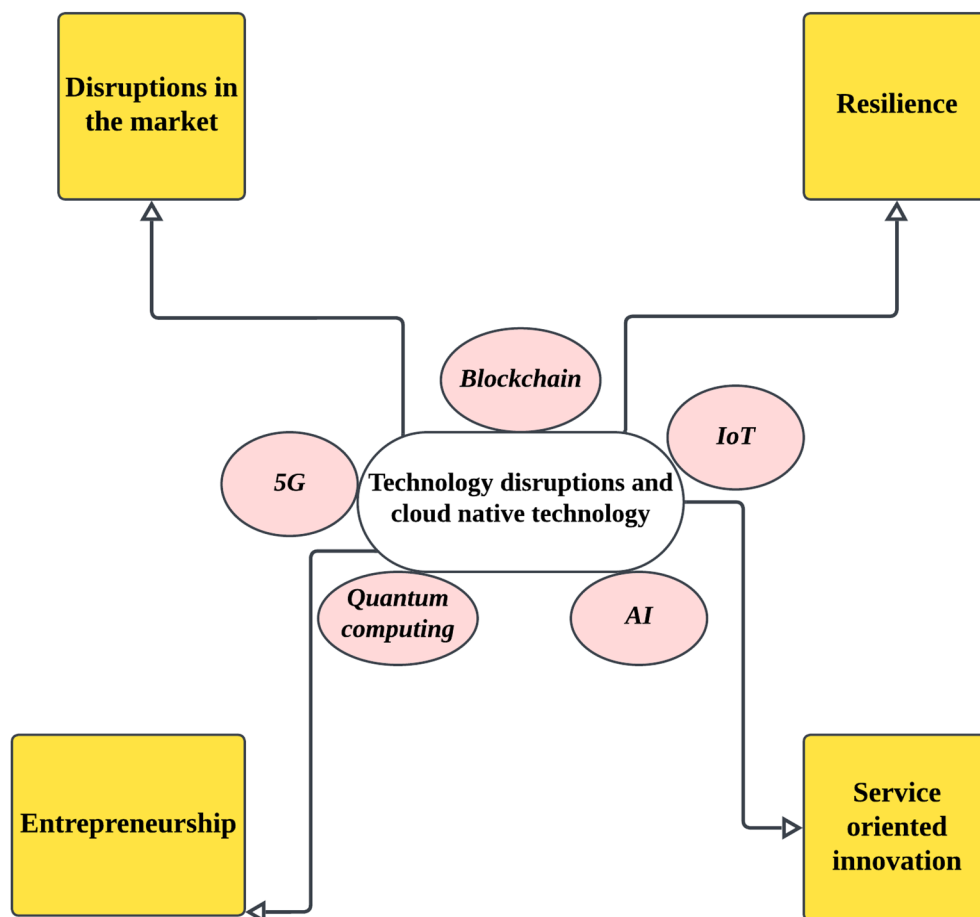


Fig. 3 Trends in cloud-native technology

as AWS Lambda and Azure Functions, are being used by entrepreneurs to streamline application development and lower infrastructure administration costs. With the help of containerization and Kubernetes, scalability and portability are made possible for cloud-native apps when they are deployed and managed. Entrepreneurs today prioritize strong security measures within their cloud-native solutions to protect against growing threats, making cybersecurity concerns essential. Last but not least, sustainability has gained popularity, with entrepreneurs aiming to reduce carbon footprints by making the most of cloud resources and incorporating eco-friendly practices into their cloud-native architectures. Cloud-native technology may support innovative business models in entrepreneurship such as “Software as a Service (SaaS),” “Serverless Computing,” “Managed Kubernetes Services,” “Data Analytics Platforms,” “Blockchain as a Service (BaaS),” “Edge Computing Solutions,” “API Marketplaces,” “FaaS for IoT Devices,” “AI and ML Platforms,” “Subscription-Box Services,” “Cloud-Native Gaming Services,” “Healthcare Telemedicine Platforms,” “Decentralized Finance (DeFi)” and “On-Demand Cloud Services Consulting.” “These include approaches that are subscription-based, freemium, usage-based, data-driven, and ecosystem-driven.”

Table 2 shows the cloud-native technology business models and their services. Apart from the types in Table 2, other business models deal with particular functions, e-commerce, and consumer-related models, and also the models that are industry-particular. The particular function business model is the API marketplace platform. Developers can find and incorporate the APIs into their use cases while also allowing creators to publish and make public their APIs. An example of this business model is Rapid API. Likewise, another business model is the telemedicine platform is a technological foundation, set of services, and assistance that enables videoconferencing for private, secure, HIPAA-compliant, and excellent virtual medical consultations including the example of Teladoc. The business models that are oriented with a particular industry are; cloud-native technology gaming services consisting of the example of Google Stadia which covers that no matter the gadget the player decides to use to play, they are inherently built to perform just as effectively. The e-commerce and consumer-related business models include subscription-box services contains the example of Blue Apron focusing on the boxes that are typically sent once a month. They can introduce clients to brands they have not previously purchased from or give them the chance to test out new items from their preferred brands.

Table 3 presents the cloud-native technology business model approaches. The following table illustrates

the cloud-native technology business model approaches, their uses, and examples of each approach to developing an understanding of the cloud-native technology business model approaches applicable to entrepreneurship.

Building a cloud-native entrepreneurial ecosystem

Cloud-native is a concept of software architecture and an approach to the practice of software development that fully exploits the benefits of cloud computing. It includes both technology (container, microservice, agile infrastructure, etc.) and management (DevOps, Continuous Delivery, Conway’s law, Reorganization, etc.) [15, 16]. Evolutionary game analysis on cloud providers and enterprises’ strategies for migrating to cloud-native under digital transformation. Man Sine (2015) introduced the idea of cloud-native architecture and identified numerous critical features, including microservices, self-service agile infrastructure, API-based collaboration, and antifragility. The ecosystem offers a variety of microservices that are related to many functional domains, including security, data, network, management, monitoring, and messaging [91]. The emergence of cloud-native technology has changed the entrepreneurial landscape and opened up novel options for both startups and established businesses. Building and growing systems with cloud-native concepts like microservices, containers, and serverless computing is known as cloud-native entrepreneurship. The examination of how ecosystem development fosters cloud-native entrepreneurship, highlights the value of collaboration, mentoring, and resource accessibility for prospective cloud-native entrepreneurs. Businesses have discussed how employing microservices has helped them reach advantages including faster delivery, scalability, and autonomous deployment. Building an ecosystem is essential for fostering cloud-native entrepreneurship. A paradigm shift has been brought about by cloud-native technologies, which emphasize cost-effectiveness, scalability, and agility. Entrepreneurs therefore need an ecosystem that can give them the skills, information, and connections they need to successfully negotiate the challenging cloud-native landscape. The ecosystem serves as an innovation accelerator, boosting the creation and uptake of cloud-native products. Cloud-native entrepreneurship is built on collaboration within the ecosystem. Collaboration with other startups, established businesses, and cloud providers is crucial for cloud-native firms since they frequently need to interact with different services and technologies. Collaborative ecosystems allow for resource sharing, interoperability, and knowledge exchange, all of which can dramatically lower entry barriers for aspiring entrepreneurs. For cloud-native entrepreneurs to succeed, mentoring is essential. Having access to skilled mentors can be quite beneficial because they can

Table 2 Cloud-native technology business models

Types	Business Models	Examples	Uses	References
Cloud-Based Services	Software as a Service (SaaS)	Slack	A cloud provider hosts applications and makes them online accessible to customers as part of the software-as-a-service (SaaS) distribution paradigm. Independent software vendors (ISVs) can hire a different cloud service provider to host their applications.	Saeedizade and Ashtiani [65], Gey et al. [66]
	Serverless Computing	AWS Lambda	Serverless computing helps the developers from having to handle infrastructure, allowing them to create apps more quickly. The infrastructure needed to run the code is automatically provisioned, scaled, and managed by the cloud service provider.	Bhat and Shroff [67], Sewak and Singh [68]
	Managed Kubernetes	Google Kubernetes Engine	It uses hybrid environments for deployment. Run Kubernetes in data centers and manage Kubernetes clusters and apps in hybrid settings.	Jiao et al. [69], Botez et al. 70
AI platforms	Data Analytics Platforms	Snowflake	This enables to access, combine, interact with, browse, and visualize data from several sources that a business may have.	Wu et al. [71]
	AI and ML Platforms	IBM Watson	It is computer software that imitates human cognition to complete challenging tasks and gain knowledge from them. Machine learning (ML) is a branch of AI that employs algorithms that have been trained on data to create flexible models.	Pinnaka et al. [72], Perniu et al. [73]
	Blockchain as a Service (BaaS)	Microsoft Azure Blockchain Service	It is beneficial to concentrate on the web application's front end and business logic rather than worrying about its back end's scalability, security, and dependability.	Prangon and Wu [74], Bellaj et al. [75]
Blockchain and Finance	Decentralized Finance (DeFi)	Compound Finance	A new approach for managing and facilitating financial services, exchanges, and transactions based on cryptocurrencies is known as decentralized finance (DeFi).	Alamsyah et al. [76], Harvey and Rabetti [77], Ferreira [78]
	FaaS for IoT Devices	AWS IoT Greengrass	It can be applied to the development of back-end systems or tasks like data processing, format conversion, encoding, and data aggregation.	Kulkarni et al. [79], Risco et al. [80], Vaño et al. [32]
	Edge Computing Solutions	Cloudflare Workers	Edge computing improves network traffic management, decreases data processing latency, speeds up responses, and facilitates compliance with governmental security and privacy regulations.	Alsadie [81], Chen et al. [82], Xu et al. [83]

Table 3 Cloud-native technology business model approaches

Business models approach	Uses	Examples	References
Subscription-Based	Where clients pay a monthly or annual charge to use your product or service, is simple to offer thanks to cloud-native technology. Building a devoted customer base and generating recurring revenue can both be accomplished through this.	Netflix, Spotify	Dutta and Pathak [84]
Freemium	In which customers can use a basic version of your product or service for free but must pay for additional features or capabilities, is also made simple by cloud-native technology. This can be an effective technique to draw in new clients and keep them coming back for more.	Canva Dropbox	Bhardwaj [85]
Usage-Based	Which clients pay for your product or service depending on how much they use it, are simple to offer thanks to cloud-native technologies. This can be a terrific strategy to draw in price-conscious customers and scale your firm as your clientele expands.	Amazon Web Services, Snowflake	Laatikainen and Luoma [86], Shaari et al. 87
Data-Driven	Collecting and analyzing data from your consumers and products is made simple by cloud-native technology. This information can be utilized to create new features and products, enhance consumer satisfaction, and make wiser business decisions.	Coca-Cola, Starbucks	Dhoni [88]
Ecosystem-Driven	Integrating cloud-native technology with other goods and services is simple. By doing this, you can create ecosystems around your good or service that will draw in additional clients and collaborators.	Apple, Amazon	Langen [89], Pan et al. 90

offer crucial advice and insights given the technical complexity of cloud-native technologies. Beyond technical skills, mentoring also provides advice on business tactics, networking, and overcoming obstacles unique to cloud-native businesses. Mentorship programs designed for cloud-native entrepreneurs can aid them in navigating the complexities of this developing industry. Additionally, platforms, frameworks, and tools that simplify development and deployment should be made more widely available through cloud-native ecosystems. For cloud-native entrepreneurs, having access to cloud infrastructure and technologies is a crucial prerequisite. Cloud providers provide startup credits and resource grants to lessen the early cost strain and enable startups to experiment and scale quickly. Cloud computing is a key enabler for tech start-ups across Asia and the Pacific [92]. One of the most important aspects of cloud-native entrepreneurship is securing financial resources. Entrepreneurs' chances of obtaining finance might be considerably improved by ecosystems that link them to venture capitalists, angel investors, or government grants. Financial resources are available for early development, ongoing growth, and innovation [69].

Ethical and regulatory considerations

Innovative technology offers opportunities for the enterprise's growth and market expansion. However, there are challenges associated with data security, privacy, compliance, and management. Trends such as AI and ML have

ethical implications, and entrepreneurs should maintain transparency. The AI must be used wisely without making any preconceptions about a particular group [93]. The European Commission provides ethical guidelines on AI with consideration to protect human values and rights, for example, the General Data Protection Regulation (GDPR). The guidelines mainly deal with the protocols for the assembling, executing, and warehousing of the data with security measures process and also direct the entrepreneurs to adherence to the compliance [94]. Entrepreneurs are encountering issues that are supposed to answer the ethical questions and the constraints. Ethical cloud-native applications are essential which address compliance, data privacy, and ethical practices in technology usage, and it reduces the risks to all the customers as well as the stakeholders. The first ethical issue involved in cloud-native technology is the privacy, security, and breach of the data, especially the presence of microservices and containerized applications [27, 95, 96]. The accountability toward society's impact on the sound is also one of the ethical issues. The major regulatory issues in cloud-native technology are adherence and the following of the laws related to privacy and data security. Technology is evolving and developing fast but the regulations are updating at the same rate as we compared with the rate of rapid trends in technology. The urgency to incorporate and develop new security features and tools for measurement is necessary. The "zero trust" model is essential in the case of the 5G cloud-native



Fig. 4 Challenges in cloud-native technology in entrepreneurship

environments due to the principle of the "Never Trust, Always Verify" [95]. The major recommendation is for policymakers to frame and implement actionable strategies for the stakeholder's protection that address the ethical uses including privacy, security, breach of data, etc. Figure 4 illustrates challenges in cloud-native technology in entrepreneurship.

The process through which new technologies displace established markets is known as technology disruption. Cloud-native technologies are changing how software is created and used. The following are some of the newest developments in cloud-native technology and technological disruption: A cloud computing execution model known as "serverless computing" allows the cloud provider to handle server allocation and deallocation based on application demand dynamically. Because of this, developers are free to concentrate on creating and distributing applications rather than managing servers. "Edge computing" is a distributed computing paradigm that moves computation and data storage to the network's edge, or the location closest to the data sources. Applications that demand real-time processing benefit from a reduction in latency and an increase in performance. Many cloud-native development and operations processes are being automated and optimized with the help of AI and ML. Automating processes like code review, performance tweaking, and security monitoring is one example of how AI/ML is employed. An open-source container orchestration technology called "Kubernetes"

automates many labor-intensive steps required to set up and maintain containerized applications. Kubernetes is now supported by all of the main cloud providers and is growing in popularity for cloud-native development. Multi-cloud and hybrid cloud deployments are getting increasingly popular as businesses try to use the finest services from many cloud providers. Although managing multi-cloud and hybrid cloud installations can be challenging, numerous tools and technologies are available to support businesses in making this process less complicated. There are a variety of potential effects on entrepreneurship from the new trends in cloud-native technologies and technological disruption, including: by removing the need to buy and maintain their hardware and software, cloud-native technologies can assist entrepreneurs in lowering their IT costs. Entrepreneurs that use cloud-native solutions will be more adaptable and receptive to market developments. For instance, cloud-native applications may be quickly scaled up or down to satisfy changing demand. By giving them access to various cloud services and tools, cloud-native technology can aid entrepreneurs in enhancing their capacity for creativity. For instance, entrepreneurs can employ cloud-based AI and ML technologies to create brand-new goods and services or enhance the performance of their current ones. Emerging technologies like edge computing and AI/ML give entrepreneurs access to new markets. For instance, entrepreneurs can create brand-new goods and services that use edge computing to enable real-time

data processing or employ AI/ML to enhance application performance. To stay ahead, entrepreneurs must keep up with the most recent developments in cloud-native technologies and technological disruption. By reading industry blogs and articles, attending industry events, and networking with other entrepreneurs and tech professionals, entrepreneurs may keep up with industry news. Opportunities for entrepreneurs to keep ahead of the curve include: Entrepreneurs that adopt new cloud-native technology early can benefit from a competitive edge. For example, early users of serverless computing have been able to lower their IT expenses considerably. Emerging technologies like edge computing and AI/ML give entrepreneurs access to new markets. Entrepreneurs might create new goods and services that use these technologies to satisfy their customers' wants. Cloud service providers provide a wide range of tools and services that can aid entrepreneurs in creating and deploying cloud-native apps. To access these tools and services and receive professional assistance, entrepreneurs can collaborate with cloud providers. Entrepreneurs may get a competitive edge and set up their companies for success in future by staying on the cutting edge and implementing the newest cloud-native technologies.

Discussion

Main findings

The cloud-native technology helps enterprises and startups with innovations, efficiency, cost-effectiveness, and flexible solutions [15, 16, 67]. The study supports these findings by answering the second research question on examining the importance of SaaS, serverless computing, managed Kubernetes services, etc. Digital transformation is essential for enterprises to remain competitive in the dynamic business environment [15, 16]. The first research question of this research analyzes the role of technology disruption and cloud-native technology in reshaping the entrepreneurial landscape. The study proves the impact of digital transformation in the technology disruptions lens. Digital transformation transforms the entrepreneurial landscape. The study supports the findings of Zhang et al. [15, 16] through the exploration of the technology disruptions and the cloud-native technology in the enterprises. Breivold and Rizvanovic [97] identified the role of service-oriented models with cloud solutions helps enterprises shift their focus from product-based development to service-based development. The study adds these findings by understanding the business model approaches of subscription-based, free-mium, and usage-based models that are emerging due to cloud-native technology. Likewise, the study by Dadawal et al. 98 explores the business models which contain the combination of AI, and blockchain, and also cud promote

the entrepreneurs to steer the dynamic digital environment. This research identifies the IoT, AI, blockchain, edge computing, etc., which are impacted in the business models and offer unique opportunities to the entrepreneurs. Bhat and Shroff 67 found that the rearrangement of the existing models. Here, this research identified the transition toward service-oriented business models and changing the entrepreneur's focus toward innovative solutions. Ali et al. [99] examined the risk concerns. In this research, the discussion on the ethical and regulatory challenges extends to the issues connected with data privacy, regulatory compliance, and recommendations for policy actions through the third research question.

The study extends the understanding to the global markets and cross-border reach of enterprises with cloud-native technology and technology disruptions. This does not limit the understanding that entrepreneurs can utilize the ecosystems and resources for their development and innovations in a holistic view of the studies [67, 97]. In a nutshell, while reviewing existing studies, they do not focus on all the business models together related to cloud-native technology, and also they are looking at the side of the advantages and disadvantages. The study discussed in terms of BaaS, serverless computing, and other new innovative business models. Kumar et al. [17] examined the technology combination that helps in the real-time decision-making and optimization of workflow. However, this study evaluated the cloud-native technology role including IoT-enabled healthcare platforms and industry-specific applications. The barriers to cloud-native technology are scaling and microservices complexity [26, 100]. However, the study identifies additional challenges of scaling by related compliance, and the keep up better performance. The solution to resolving the barriers associated with microservices is the well-developed strategies for automation. Zhang et al. [15, 16] call for the need for established monitoring techniques to evaluate the issues connected with performance. The ethical considerations are not discussed. This research calls for regulatory and policy support for effective mechanism systems, to solve the data privacy, confidentiality, and security challenges. The lean startup approach provides entry points for innovation opportunities for entrepreneurs [101]. This examination adds to this finding by understanding the collaboration and networking importance for overcoming the problems of the cost constraints to develop flexible, affordable, innovative solutions for surviving in the competitive landscape.

Practical implications

The study identifies the role of cloud-native technologies and the technology disruptions in entrepreneurship. A long-term investment is essential for installation and

the adaptability of disruptive technologies. The startups that are exclusively aimed at rapid development and focus on innovative technologies should prioritize the capitalization of the investment from different sources before the establishment of the ventures that help to generate innovative and novel solutions for the customers. The rethinking of existing business models is essential to sustain in the competitive landscape. Entrepreneurs must possess the agile mindset to exploit the technology trends such as the blockchain and cloud-native technologies to market entry as well as the innovative products that existing competitors. The existing enterprises need to invest in the new digital technologies to enjoy the benefits including cloud native. The SaaS, serverless computing, microservices, blockchain facilities, and AI/ML are promoters the flexible and growing solutions that mainly resulting in less infrastructure cost. The policymakers and institutions should promote supporting assistance to the startups and MSMEs to adopt these systems which are reshaping the development. The cloud-native technologies impact on startups is examined in this study. The study offers the idea that startups can use cloud-native technologies for innovative business models that rely on serverless computing, subscription services, and on-demand cloud services which are providing opportunities. The findings of the study highlight the need for entrepreneurs to utilize emerging technologies to meet competition, and there is a requirement for continuous investment in data analytics, automation, cloud solutions, and emerging technologies such as biotechnology, 5G, IoT, quantum computing, and AI. Collaboration is essential for the application of the technologies for early stage startups and enterprises. Incubators, accelerators, and venture capital firms support essential to the adoption of disruptive technologies. The mechanisms to solve the barriers of regulatory compliance, lack of skilled manpower, data security, privacy issues, and upfront investment are essential for entrepreneurial success. The policymakers and enterprises focus and actionable intervention is essential. The adoption of containerization and Kubernetes helps in the scalability and adaptability. It enhances the reduction of operational costs and the stimulation of growth. The entrepreneurs pay attention to environmentally friendly cloud architectures to foster sustainable business activities. The study identifies that low-end disruption creates a positive impact. The integration with conventional models and the new modern disruptive technologies will boost competitiveness for entrepreneurs.

Theoretical implications

The study is a systematic exploration of the role of technology disruption and cloud-native technology in

entrepreneurship act as catalysts for the group. The findings of the study contribute to the existing body of knowledge on both entrepreneurship and technology disruption including cloud-native technology research. The exploration of technology trends is the most interesting field in the research landscape. The study provides further research directions to the existing researchers and also the new researchers who are aspiring to understand this phenomenon by releasing the current state of the art of the literature. The study identifies the importance of cloud-native technology in entrepreneurship. Future research is required to exclusively focus on how the scalability and flexibility of enterprises are determined by cloud-native technology. Likewise, the impact and the contribution of cloud-native technology for resilience and sustainability, especially in the areas of the supply chain and the market disruptions, are another underexplored area. Global trends such as moving toward sustainable development and achievement of sustainable development goals are another potential avenue for further research. The mapping of sustainable development goals with an understanding of the direct contribution of cloud-native technology-driven entrepreneurship is most relevant in the current scenario. The impact of cloud-native technology in the circular economy in entrepreneurship and also evaluating the role of environmental impact. The study discussed the patterns and challenges of cloud-native technology in entrepreneurship. The discussion on how the challenges and the patterns significantly differ in developing and developed countries is an area that requires further examination. This research identifies that infrastructure cost is critical in their adoption. The study does not address the role of cloud-native technology in inclusivity especially in the underprivileged community, and the case of women entrepreneurs and startups focus in rural settings. The case studies and longitudinal studies explore the conventional industries' adoption of cloud-native technology focusing the serverless computing, and the SaaS remains unexplored areas. The significance of the cloud native in the entrepreneur's support mechanisms such as incubators, accelerators, and venture capital platforms is another area of further interest required. The data privacy, security, protection, and ethical aspects need to be ensured while adopting cloud-native technology in entrepreneurship. The strategies to manage the ethical issues, the investigation of the effectiveness of regulatory mechanisms, and the compliance requirements need to be discussed in future. AI most sensation nowadays. The AI-powered cloud-native technology in innovation and startup efficiency is one of the further exploration possibilities. The employees' adaptation to the changes brought out by the enterprises is crucial for their success. The impact of the cloud-native

technology application on the employee work culture, collaborative work, efficiency, and the selection of new talents as well the retention strategies in the startups and firms are need to be explored. Future research must call for the actionable intervention of the policymakers, practitioners, and entrepreneurs to utilize the cloud-native opportunity with effective management of barriers.

Recommendations

The study examines and identifies the entrepreneurship need to utilize technology trends such as cloud-native technology which promotes entrepreneurship and the opportunity to offer innovation, satisfy market demand, and also bring agility to their operations. The strategies must be adopted by the entrepreneurs to adopt these technologies to sustain in the competitive marketplace. The major strategy is to focus on the low-end disruptions that create impacts on the services and also the novel solutions. The introduction of fair, affordable, products with the help of SaaS and serverless computing can foster the startup's entry into the market. Another important strategy is the incorporation of the SaaS and AI/ML platforms for the application of subscription-based free-mium-based or usage-based pricing models. Likewise, the integration of the techniques that influence the value creation through business models the API marketplaces, or Blockchain as a Service, and the IoT, AI, and quantum computing.

The whole application is divided into microservices which are focusing each unique operation and function to create a novel solution that aligns with consumer preference, and it also reduces the difficulties of scaling the particular services. Kubernetes and Docker will help in this case to reduce the disruptions. The incorporation of serverless computing including AWS Lambda will promote smooth operations without the need to take care of the servers. This strategy will lead to reducing the cost, time, and dedicated specific operations such as authentication and APIs during the traffic and also an opportunity to focus on the main entrepreneurial activity. The development of API platforms for the integration of stakeholders, and also the third-party developers with the enterprise solutions will enhance the ecosystem. The next attention must be given to the effective management of Kubernetes which is crucial for the applications in the peak demand time, operations, and scalability. Entrepreneurs must prioritize the cloud-native technology that enables data analytics, which is essential for decision-making, especially in consumer needs, identification of requirements of marketing strategies, and also increasing the overall efficiency of the functions. The study identifies data privacy and data security as important elements in cloud-native technology. An effective measure to solve

the privacy issue and data protection through cloud-native security solutions will help in the encryption, management of identity, and adherence to regulations. The significant next focus is the integration pay-as-you-go model which manages the cost in the low demand time. The resource allocation and reduction of extra provision of the resources can be eliminated through this. The entrepreneurs must incorporate innovative effective strategies for cloud-native technology which is most prominent in the current business environment.

Conclusion

Technology disruption and cloud-native technology play a significant role in entrepreneurship. Hence, the answer to the first research question is that low-end disruption has a positive but limited effect. Manufacturing, distribution, and transaction are the three areas where digital entrepreneurs can capitalize on the old value-delivery methods being phased out by technology shocks. Digital disruptors can exploit digital platforms by producing new products, improving the usability of old products, and connecting them digitally. Disruptive technologies either increase the number of industries they enter or create whole new ones by offering goods and services that are drastically more affordable, superior, and convenient. These disruptive technologies frequently harm how people participate in labor because they enable those with low technological skills to compete and enter the workforce. Disruptive businesses bring in products and services that are drastically more affordable, superior, and practical. One of the main forces behind disruption is the quick development of new technologies including "biotechnology," "blockchain," "5G," "Internet of Things (IoT)," "quantum computing," and "artificial intelligence (AI)." By developing novel products and business models, these technologies enable entrepreneurs to compete with long-standing industry leaders. The conversion of old processes to digital ones, which includes automation, data analytics, and cloud computing, opens up new avenues for business operations and customer service. Customer preference shifts, particularly in the digital trend, also fuel disruption by changing how companies engage with customers and provide goods and services. Online streaming, social networking, mobile apps, and e-commerce have completely changed this market. Strong startup ecosystems with access to accelerators, incubators, and venture financing offer a favorable environment for disruption, enabling creative businesspeople to take on market leaders. Most current trends in entrepreneurs' usage of cloud-native technology include an increasing emphasis on cybersecurity within cloud-native designs, increased acceptance of serverless computing and Kubernetes, and the creation of environmentally friendly

cloud-native solutions. Entrepreneurs employ serverless technologies like “AWS Lambda” to streamline application development and reduce the cost of maintaining infrastructure. Scalability and mobility are made possible for cloud-native apps when they are deployed and managed with the aid of containerization and Kubernetes.

The answer to the research question is that cloud-native technology supports innovative business models in entrepreneurship such as “Software as a Service (SaaS),” “Serverless Computing,” “Managed Kubernetes Services,” “Data Analytics Platforms,” “Blockchain as a Service (BaaS),” “Edge Computing Solutions,” “API Marketplaces,” “FaaS for IoT Devices,” “AI and ML Platforms,” “Subscription-Box Services,” “Cloud-Native Gaming Services,” “Healthcare Telemedicine Platforms,” “Decentralized Finance (DeFi)” and “On-Demand Cloud Services Consulting.” These include approaches that are subscription-based, freemium, usage-based, data-driven, and ecosystem-driven. Regulatory changes can disrupt a sector by imposing limits or opening up new opportunities. These changes will be advantageous to entrepreneurs who understand the subtleties of the regulatory environment. The connected world economy enables quick cross-border expansion and supports disruptive businesses by providing access to global markets, talent, and resources. Entrepreneurs can access external knowledge, technologies, and resources through collaborative ecosystems and open innovation methodologies, accelerating innovation and undermining conventional business paradigms. The popularity of sustainable and socially conscious enterprises has increased with the growing awareness of environmental and social concerns. The advancement of security techniques and technology is being driven by challenges about cybersecurity and data privacy, which is increasing demand for entrepreneurs who can offer trustworthy cybersecurity solutions. Advanced technological tools and platforms, such as low-code/no-code platforms, cloud computing, and open-source software, have become more accessible, enabling entrepreneurs with limited resources to launch and grow their firms. The economy, remote work, and changing labor standards are redefining team dynamics and company processes.

Applying disruptive technology creates challenges for entrepreneurs including the requirement for investments in new technologies and infrastructure, regulatory issues and compliance problems, the potential for market uncertainties and unpredictable customer adoption rates, and the requirement for qualified talent to successfully implement and manage the new technology. Furthermore, safeguarding data security and privacy in applications of disruptive technology can be challenging, particularly when dealing with sensitive data. Even

so, conquering them can lead to significant rewards and a competitive edge in the ever-evolving technological environment. Ethical cloud-native applications are essential which address compliance, data privacy, and ethical practices in technology usage, and it reduces the risks to all the customers as well as the stakeholders. The first ethical issue involved in cloud-native technology is the privacy, security, and breach of the data, especially the presence of microservices and containerized applications. The major regulatory issues in cloud-native technology are adherence and the following of the laws related to privacy and data security. Technology is evolving and developing fast, but the regulations are updating at the same rate as we compared with the rate of rapid trends in technology. The urgency to incorporate and develop new security features and tools for measurement is necessary. The major recommendation is for policymakers to frame and implement actionable strategies for the stakeholder's protection that address the ethical uses including privacy, security, breach of data, etc.

The major limitation of the study is that this research is a systematic literature review. Therefore, the firsthand data examination is missing. The major aim of this study is to evaluate the current art of the existing literature. The study was conducted last October 2024. The result is not constant. Future study is needed to examine the body of knowledge over the period. This follows the inclusion and exclusion criteria. If not limited to the criteria such as the selection of documents in the English language and papers below the quality of conference proceedings, then the results will change. One of the shortcomings experienced while doing the study was the smaller number of studies. This limitation denotes that there is scope for different studies in the form of case studies and longitudinal studies regarding this topic in the coming years. The inherent limitations of applying this type of literature review with the PRISMA methodology also affected the result. The non-application of primary data lacks firsthand information from the entrepreneur's perspective.

The findings of this study offer directions and knowledge on technology disruption and cloud-native technology applications in entrepreneurship. Hence, it can be utilized by entrepreneurs, researchers, and other interested parties as a model or a kind of source of information for better understanding and also helps in identifying the recent trends in this theme. The study offers directions to the startups that are exclusively aimed at rapid development and focus on innovative technologies that should prioritize the capitalization of the investment from different sources before the establishment of the ventures that help to generate innovative and novel solutions for the customers. The rethinking of existing business models is essential to

sustain in the competitive landscape. Entrepreneurs must possess the agile mindset to exploit the technology trends such as the blockchain and cloud-native technologies to market entry as well as the innovative products that existing competitors. The policymakers and institutions should promote supporting assistance to the startups and MSMEs to adopt these systems which are reshaping the development. The findings of the study highlight the need for entrepreneurs to utilize emerging technologies to meet competition, and there is a requirement for continuous investment in data analytics, automation, cloud solutions, and emerging technologies such as biotechnology, 5G, IoT, quantum computing, and AI. The mechanisms to solve the barriers of regulatory compliance, lack of skilled manpower, data security, privacy issues, and upfront investment are essential for entrepreneurial success. The policymakers and enterprises focus and actionable intervention is essential.

The findings of the study contribute to the existing body of knowledge on both entrepreneurship and technology disruption including cloud-native technology research. The study provides further research directions to the existing researchers and also the new researchers who are aspiring to understand this phenomenon by releasing the current state of the art of the literature. Future research is required to exclusively focus on how the scalability and flexibility of enterprises are determined by cloud-native technology; the impact and the contribution of cloud-native technology for resilience and sustainability, especially in the areas of the supply chain and market disruptions, is another underexplored area; the challenges and the patterns significantly differ in developing and developed countries. This is also one of the potential avenues. The study is important due to the comprehensive examination of cloud-native technology in reshaping the entrepreneurial landscape. The identification of the cloud-native technology-enabled business models, approaches, challenges, and ethical barriers and recommendations to solve the dilemmas is important. The snapshot of the intersection of the technology and the entrepreneurship field shows the potential impact of the cloud-native technology capable of crucial driver for transformation.

Acknowledgements

Authors are thankful to the anonymous reviewer's suggestion to improve this manuscript quality.

Author contributions

T.A. Alka is the first author, contributed to conceptualization and prepared original manuscript; Aswathy Sreenivasan is the second author and contributed to data analysis; visualization; and manuscript editing; and M. Suresh is the third author and contributed to final manuscript preparation and supervision.

Funding

No funding support received for this research work

Availability of data and materials

Availability of data in the Manuscript.

Code availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

All authors have declared that there are no competing interests regarding financial and non-financial.

Received: 14 February 2024 Accepted: 14 January 2025

Published online: 05 February 2025

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Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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